NED University of Engineering and Technology
Department of Computer Science and Information Technology

Course: Data Mining (CT-377)

CCP Project Report

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1. Introduction:

This project implements a keyword-driven pipeline that collects, processes, and analyses blog posts from multiple web sources using data mining and NLP to surface insights about content tone, authorial intent, and reader sentiment. It surfaces encyclopedic, tutorial, and discussion content across all three sources, yielding a heterogeneous corpus with strong signal for classification and sentiment analysis. Ten articles were collected from **Wikipedia**, **Dev.to**, and **HackerNews**, processed through TF-IDF and LDA analysis, enriched with real scraped and AI-generated comments, and visualized across five charts and one word cloud.

2. Methodology:

2.1 Data Collection:

Source	Access Method	Articles
Wikipedia	HTML scrape of 5 keyword-variant URLs	Up to 5
Dev.to	REST API	Up to 6
HackerNews	Algolia API + nested JSON	Up to 8

2.2 Preprocessing and NLP Analysis:

Raw text is cleaned via: lowercasing -> newline removal ->punctuation stripping -> NLTK stopword removal (179 tokens) -> truncation to 2,000 characters. The cleaned **clean_text** column feeds all NLP steps. The original **raw_text** is preserved for motive and tone detection, which rely on natural phrasing.

TF-IDF Keyword Extraction: TfidfVectorizer (max_features = 50) extracts the five highest-scoring terms per article, penalizing corpus-wide frequent terms in favour of discriminating per-document tokens.

Topic Modeling (LDA): Applied to the TF-IDF matrix with n_components = min(3, len(df)). Three topics extracted, each represented by five keywords. random_state = 42 ensures reproducibility.

Motive Inference: Rule-based classifier using trigger phrases — Tutorial ("how to", "guide"), Promotional ("buy", "pricing"), Academic ("research", "study"), Opinion ("i think", "i believe"), Informative (default).

Tone Detection: TextBlob .sentiment.subjectivity > 0.5 -> **Subjective**; ≤ 0.5 → **Objective**.

2.3 LLM Comment Generation and Sentiment Analysis:

Two models were evaluated for personalized comment generation:

- **GPT-2** is a next-token predictor. It continues the prompt text rather than following it as an instruction.

- **google/flan-t5-base**, fine-tuned on 60+ instruction tasks, genuinely interprets the prompt and generates contextually relevant comments without an API key.

Sentiment classification uses **TextBlob** polarity on two channels:

- **Scraped**: They are the real user comments from Dev.to and HackerNews APIs.

- **Generated**: It is the Flan-T5 output.
  Thresholds: > 0.1 → Positive, < −0.1 → Negative, else Neutral.
  A popularity proxy **comment_count = int(|polarity| × 100) + random(5, 20)** is used in the scatter plot.

3. Results and Findings:

Assuming that we run our pipeline with keyword = “Machine Learning”. Hence, we obtain the following results:

3.1 Collected Articles and Topic Modelling:

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──────────────────────────────── Webscraping & NLP Pipeline ────────────────────────────────

Keyword: Machine Learning
Using keyword: Machine Learning

──────────────────────────────── Step 1 — Data Collection ───────────────────────────────
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Source	Articles
Wikipedia	1
Dev.to	0
HackerNews	8

```
Fetching real user comments from Dev.to and HackerNews...
Real comments fetched for 8 article(s)
```

Collected Articles [10 total]

#	Title	Author	Date	Source
1	Machine learning	Wikipedia Contri…	2024-01-01	Wikipedia
2	Machine Learning Crash Course	matant	2018-03-01	HackerNews
3	Machine Learning 101 slidedeck: 2 years …	flor1s	2017-12-14	HackerNews
4	The Chaostron: An Important Advance in L…	abrax3141	2022-10-15	HackerNews
5	Apple's director of machine learning res…	carlycue	2022-05-07	HackerNews
6	Ask HN: How to get started with machine …	georgehdd	2016-10-15	HackerNews
7	Distill: a modern machine learning journ…	jasikpark	2017-03-20	HackerNews
8	Math Basics for Computer Science and Mac…	oldgun	2019-07-30	HackerNews
9	Google supercharges machine learning tas…	hurrycane	2016-05-18	HackerNews
10	Introduction to Machine Learning	John Doe	2024-01-01	Backup

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──────────────────────────────── Step 2 — NLP Analysis ──────────────────────────────────
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LDA Topics

Topic	Top Keywords
Topic 1	applications, google, cover, modules, years
Topic 2	think, framework, tensorflow, software, started
Topic 3	computer, field, pdf, machine, learning

NLP Per-Article Results

Title	Source	Motive	Tone	Keywords
Machine learning	Wikip…	Acade…	Objec…	learning, algorithms, machine, human, mathematical
Machine Learning Crash Course	Hacke…	Infor…	Objec…	cover, modules, best, building, models
Machine Learning 101 slidedec…	Hacke…	Infor…	Objec…	years, machine, learning, 27, ago
The Chaostron: An Important A…	Hacke…	Infor…	Subje…	learning, 27, ago, algorithms, apple
Apple's director of machine l…	Hacke…	Tutor…	Objec…	days, apple, employees, ago, new
Ask HN: How to get started wi…	Hacke…	Infor…	Objec…	started, framework, software, tensorflow, think
Distill: a modern machine lea…	Hacke…	Acade…	Subje…	new, ideas, work, research, community
Math Basics for Computer Scie…	Hacke…	Infor…	Objec…	computer, pdf, machine, learning, 27
Google supercharges machine l…	Hacke…	Acade…	Subje…	google, applications, tpus, machine, tpu
Introduction to Machine Learn…	Backup	Infor…	Subje…	field, intelligent, machine, learning, 27

3.2 Motives, Tones and Generated Comments:

Tone by Source

Source	Tone	Count
Backup	Subjective	1
HackerNews	Objective	5
HackerNews	Subjective	3
Wikipedia	Objective	1

#	Title	Comment
1	Machine learning	In a world where machine learning is becoming more and
2	Machine Learning Crash Course	The Crash Course: Machine Learning Crash Course is a Cr
3	Machine Learning 101 slidedeck: 2…	You're not the only one who's been trying to learn how
4	The Chaostron: An Important Advan…	The Chaostron: An Important Advance in Learning Machine
5	Apple's director of machine learn…	Apple's director of machine learning, Steve Jobs, resig
6	Ask HN: How to get started with m…	In a recent blog post, HN.com explains how to get start
7	Distill: a modern machine learnin…	MIT's Distill: A Modern Machine Learning Journal is a j
8	Math Basics for Computer Science …	In Math Basics for Computer Science and Machine Learnin
9	Google supercharges machine learn…	Google's TPU custom chip is a supercharger for machine
…	Introduction to Machine Learning	This is a great introduction to machine learning, and i

3.3 Sentiment Analysis and Visualizations:

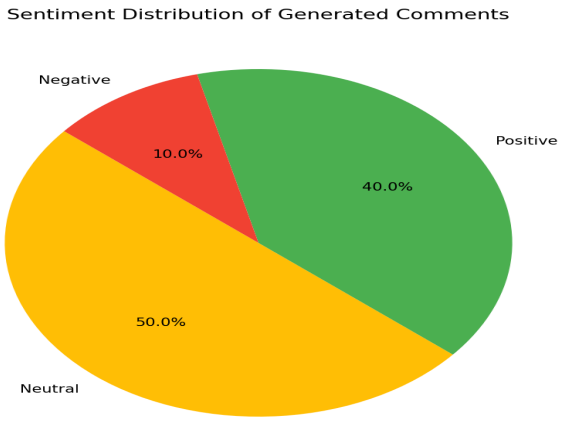
Step 4 — Sentiment Analysis

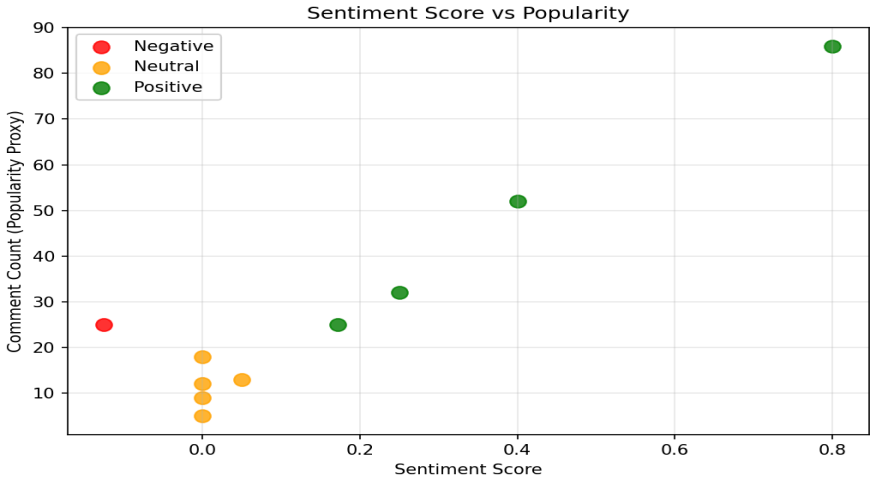
Sentiment Results

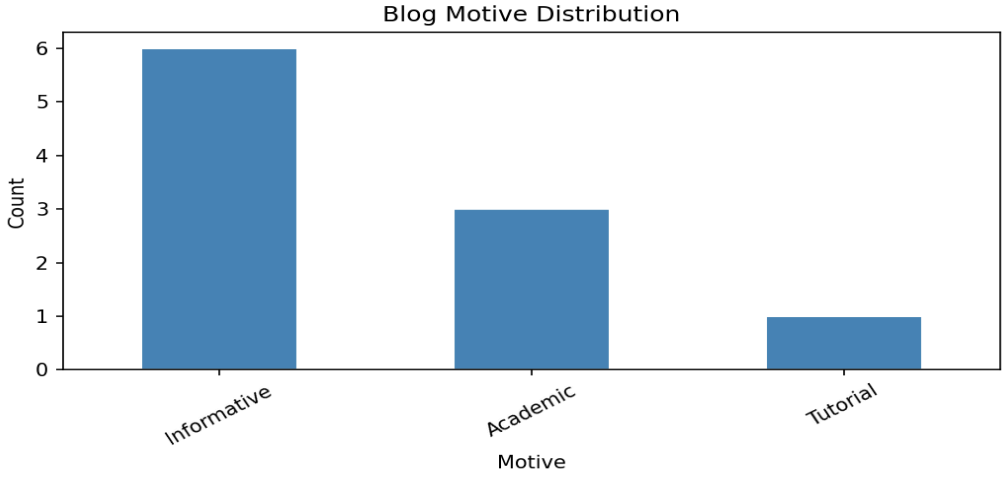
#	Title	Scraped Sent.	Generated Sent.	Score	Populari…
1	Machine learning	Neutral	Positive	0.17	25
2	Machine Learning Crash Course	Positive	Neutral	0.00	5
3	Machine Learning 101 slidedeck: …	Neutral	Negative	-0.12	25
4	The Chaostron: An Important Adva…	Neutral	Positive	0.40	52
5	Apple's director of machine lear…	Neutral	Neutral	0.00	12
6	Ask HN: How to get started with …	Positive	Neutral	0.00	9
7	Distill: a modern machine learni…	Positive	Neutral	0.05	13
8	Math Basics for Computer Science…	Positive	Positive	0.25	32
9	Google supercharges machine lear…	Neutral	Neutral	0.00	18
10	Introduction to Machine Learning	Positive	Positive	0.80	86

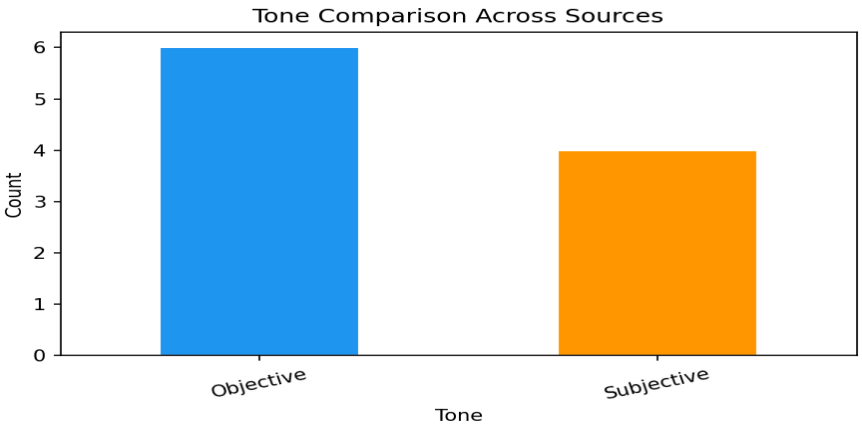
Sentiment Distribution

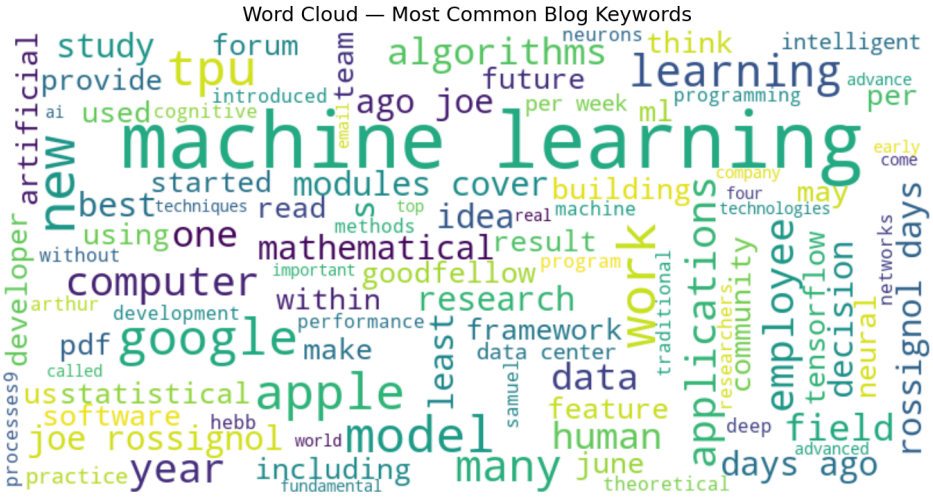
Type	Positive	Neutral	Negative
Scraped	5	5	0
Generated	4	5	1


Sentiment Distribution of Generated Comments
Negative
10.0%
Positive
40.0%
50.0%
Neutral


Sentiment Score vs Popularity
Negative
Neutral
Positive
Comment Count (Popularity Proxy)
90
80
70
60
50
40
30
20
10
0.0
0.2
0.4
0.6
0.8
Sentiment Score


Blog Motive Distribution
Count
6
5
4
3
2
1
0
Informative
Academic
Tutorial
Motive


Tone Comparison Across Sources
Count
0
1
2
3
4
5
6
Objective
Subjective
Tone


Word Cloud — Most Common Blog Keywords
machine learning
learning
algorithms
tpu
google
apple
model
many
year
computer
mathematical
research
framework
data
human
field
employee
applications
work
rossignol days
decision
tensorflow
neural
study
forum
provide
artificial
new
best
started
modules
cover
building
idea
one
using
goodfellow
within
pdf
make
least
feature
joe rossignol
usstatistical
software
including
june
days
ago
future
ago joe
per week
ml
programming
intelligent
think
neurons
advance
per
early
come
company
four
may
technologies
machine
real
top
read
techniques
methods
result
program
important
without
developer
arthur
development
performance
data center
traditional
researchers
community
networks
neural
deep
advanced
theoretical
fundamental
practice
processes9
called
hebb
world
samuel
ai
used
cognitive
introduced
email
team

4. Conclusion:

Ten articles are collected from Wikipedia, Dev.to, and HackerNews via authentication-free APIs bypassing the bot-detection barriers that blocked all mainstream alternatives and then processed through TF-IDF keyword extraction, LDA topic modeling, rule-based motive inference, and tone classification. Sentiment is evaluated on two parallel channels: real scraped user comments and Flan-T5-generated personalized comments, producing a dual-channel view that most implementations do not provide. Results are communicated through five charts and one word cloud. The key technical contribution is substituting GPT-2 with google/flan-t5-base.

5. Future Improvements:

- Upgrading sentiment modelling for noisy social text by comparing TextBlob with a stronger model (e.g., VADER or RoBERTa-based sentiment) and then performing report validation on a small manually labeled sample.

- Adding source-wise quality checks (minimum text length, duplicate removal, language filter), log failed fetches, and fix seeds/configs for deterministic runs.